Fertility, Housing, and Population Dynamics

From a stationary closure to a dated transition

Tommaso De Santo

August 2026

Equilibrium closure and transition dynamics

Last time: discussion of how to close population and housing markets:

- closed versus open population, and the role of migration and household formation;
- stationary equilibrium versus a dated transition;
- fertility as a source of future household demand.

This time: consider a closed economy that begins in a steady state and experiences a permanent fall in the preference for children:

- fertility and the house price respond on impact;
- cohort sizes then change population and housing demand;
- the path may converge to a new steady state;
- closed and open population versions are then compared in the quantitative model.

A toy economy

To fix ideas, consider a simplified economy with two household generations:

- At date t, there are Y_t young households and O_t old households; total population is $S_t = Y_t + O_t$.
- A household lives for two periods. A young household becomes old next period; an old household exits after the current period.
- Young households choose consumption, adult housing, and n_t **household children**. Each household child becomes one young household next period.
- Old households choose consumption and housing but not fertility.
- Both generations purchase housing services in one competitive market at price p_t.

There is no migration or asset accumulation. The timing implies

$$(Y_t, O_t) \xrightarrow{\text{choices at } p_t} (Y_{t+1}, O_{t+1}) = (n_t Y_t, Y_t).$$

The quantitative model restores age, tenure, mortgages, saving, earnings risk, and bequests.

Household problems

A young household earns y_Y and chooses consumption c_Y, adult housing h^A, and household children n:

$$\max_{c_Y, h^A, n} \left\{ c_Y + \alpha \log h^A + v \log n \right\}, \qquad c_Y + ph^A + \big[q + a(p+\omega)\big] n = y_Y.$$

The parameter α values adult housing and v values children. Each household child requires q units of other goods and a units of housing; ω is the extra unit cost of child-suitable housing. Fertility and total young-household housing demand are

$$n(p; v) = \frac{v}{q + a(p+\omega)}, \qquad h_Y(p; v) = \frac{\alpha}{p} + an(p; v).$$

An old household earns y_O and has housing weight $\bar{h}_O$:

$$\max_{c_O, h_O} \left\{ c_O + \bar{h}_O \log h_O \right\}, \qquad c_O + ph_O = y_O,$$

which gives $h_O(p) = \bar{h}_O / p$. Consumption is the numeraire, and incomes are high enough for the interior choices shown above.

A higher housing cost lowers fertility and housing demand.

Competitive equilibrium

Housing is supplied competitively according to

$$H^s(p) = \bar{H} p^{\varepsilon},$$

where $\bar{H}$ is the supply scale and ε is the price elasticity.
Given (Y_0, O_0) and a path for child preferences $\{v_t\}$, an equilibrium is a sequence of prices, household choices, and cohort masses satisfying

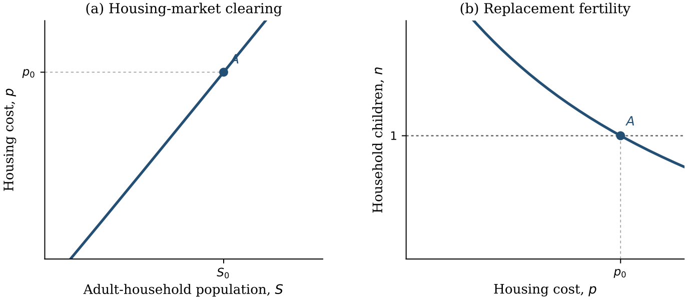
$$Y_t h_Y(p_t; v_t) + O_t h_O(p_t) = H^s(p_t),$$

and

$$Y_{t+1} = n(p_t; v_t) Y_t, \qquad O_{t+1} = Y_t.$$

Current cohort masses determine the market-clearing price; the price determines fertility and therefore the next cohort.

Initial steady state



A steady state is a constant triple (Y^*, O^*, p^*). It requires

$$n(p^*; v_0) = 1, \qquad Y^* = O^*.$$

Replacement selects

$$p_0 = \frac{v_0 - q}{a} - \omega.$$

Housing clearing then selects $S_0 = S^*(p_0)$, where

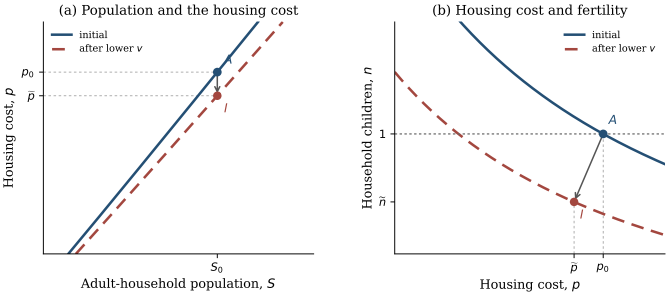
$$S^*(p) = \frac{2\bar{H}p^{1+\varepsilon}}{\alpha + \bar{h}_O + ap}.$$

Point A is the initial steady state.

Along the steady-state schedule, direct differentiation gives

$$\frac{dS^*}{dp} = \frac{2\bar{H}p^{\varepsilon}\left[(1+\varepsilon)(\alpha + \bar{h}_O) + \varepsilon ap\right]}{(\alpha + \bar{h}_O + ap)^2} > 0 \qquad \text{for } \varepsilon \geq 0.$$

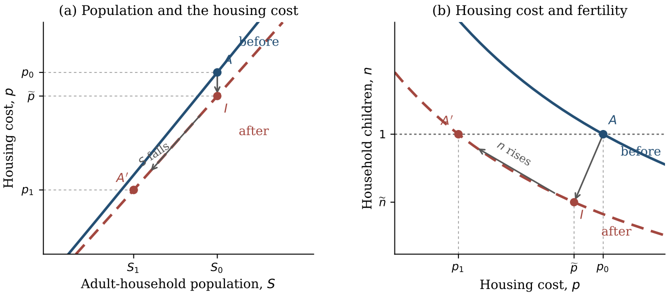
Impact response



- **Shock:** $v_0 \to v_1 < v_0$.
- **Inherited cohorts:** $Y_t = Y_0$, $O_t = O_0$, $S_t = S_0$.
- **Housing cost:** lower child-related demand gives $p_0 \to \widetilde{p}$.
- **Fertility:** $\widetilde{n} = n(\widetilde{p}; v_1) < 1$.
- **Point I:** the impact equilibrium with inherited cohort sizes.

Demographic adjustment

▸ Dynamic path



- **Below replacement at I:**

$$Y_{t+1} = \widetilde{n} Y_t < Y_t.$$

- **Cohorts:** smaller young cohorts eventually reduce S.
- **Housing cost:** lower demand moves $\widetilde{p} \to p_1$.
- **Fertility:** lower p raises n along the red curve.
- **Point A':** $n = 1$, $p_1 < p_0$, and $S_1 < S_0$

Lower prices cushion the contraction if the fertility schedule crosses replacement at a positive population level.

Below-replacement fertility and population dynamics

Let $n_t = n(p_t; v_t)$ be household children per young household and $\mu_t = O_t/Y_t$ the old-to-young cohort ratio. Population can rise with $n_t < 1$ whenever

$$\mu_t < n_t < 1.$$

A numerical example

$$100 \text{ young} + 70 \text{ old} \quad \longrightarrow \quad 80 \text{ young} + 100 \text{ old}$$

$n_t = 0.8 < 1$, but total households rise from 170 to 180.

Housing costs can rise as well if the aging cohort retains more housing than is released by the shrinking young cohort:

$$(1 - \mu_t)h_O(p_t) > (1 - n_t)h_Y(p_t).$$

Inherited cohorts can sustain short-run population and housing-cost growth below replacement.

Quantitative options

Exercise	**Exogenous demographic input**	**Endogenous outcomes**	**Interpretation**
2007–2023 bridge	Household totals and age shares at each date	Prices and household choices within each age group	Historical accounting and 2023 calibration
Semi-open historical path	Four age-group residual flows	Cohort evolution around those flows	Requires a separate household-formation margin
Closed post-2023 path	The fitted 2023 state; no outside entry	Births, cohort sizes, prices, and tenure after 2023	National finite-horizon benchmark
Open post-2023 path	A fixed outside inflow and retention rate	The same endogenous choices and prices	Entry-flow sensitivity

Main specification: closed post-2023 path. **Sensitivity:** open post-2023 path.

Dated quantitative transition

Household problem. Households choose fertility, tenure, housing, saving, and location. The distribution G_t records their joint distribution and mass.

Calendar-time accounting.

$$G_{t+1} = \mathcal{T}(G_t, p_t) + E_{t+1}, \qquad E_{t+5} = \frac{b_t^{\text{child}}}{2.1}.$$

The transition $\mathcal{T}$ moves households across age and economic states. Here b_t^{child} counts children; 2.1 children generate one future entrant household under the replacement normalization. Entrants arrive after twenty years.

Housing market. At each date, the housing cost clears demand from the current distribution:

$$H^s(p_t) = D(G_t, p_t).$$

- **2007–2023:** household totals and age shares are imposed; the model determines states within age.
- **After 2023:** cohort masses evolve under the demographic law above.

Calibration from 2007 to 2023

1. **Replacement normalization:** completed fertility equals 2.1.
2. **2007 state:** match age shares and preserve economic states conditional on age.
3. **2007–2023:** impose demographics; estimate eleven parameters from twelve moments.
4. **Validation:** compare untargeted fertility, timing, tenure, rooms, house prices, and rents.
5. **Post-2023:** hold preferences fixed; evolve the inherited distribution and entrant cohorts.

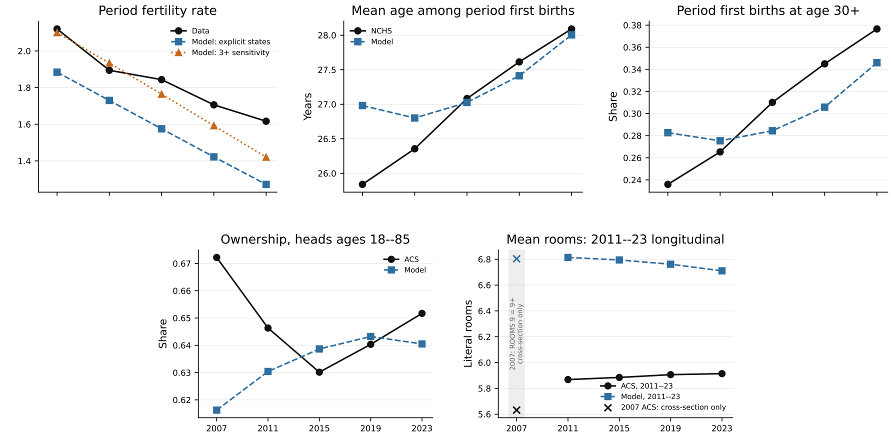
Demographic inputs: household totals and age shares. **Initial condition:** the 2007 distribution.

2023 calibration

Moment	**Target**	**Model**
Completed fertility	1.918	1.918
Childless share	0.188	0.186
Mean age at first birth	26.045	26.327
Share of first births at age 30+	0.260	0.242
Rooms response around first birth	0.720	0.380
Rooms gap: 3+ versus 1–2 children	0.368	0.359
Ownership gap: family versus no family	0.168	0.177
Ownership rate	0.575	0.546
Mean occupied rooms	5.780	6.710
Wealth / annual labor earnings	6.873	6.766
Annual bequests / aggregate wealth	0.0088	0.0086
Old-age wealth p90/p50	3.448	3.396

Largest discrepancies: the rooms response around first birth and mean occupied rooms.

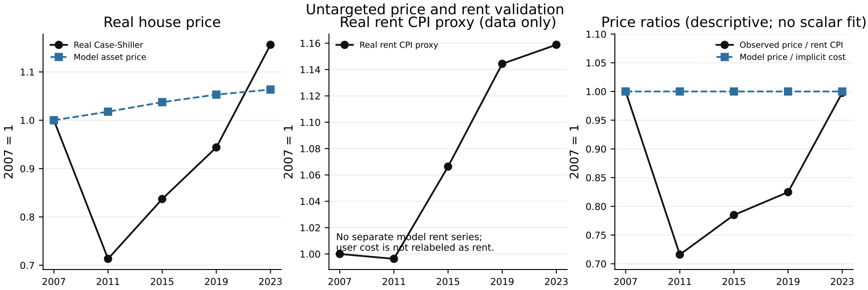
Historical validation



Birth timing is close by 2023; ownership and rooms dynamics are missed, and births per adult household fall too rapidly.

Sources: NCHS natality, World Bank period fertility, and ACS.

House-price dynamics

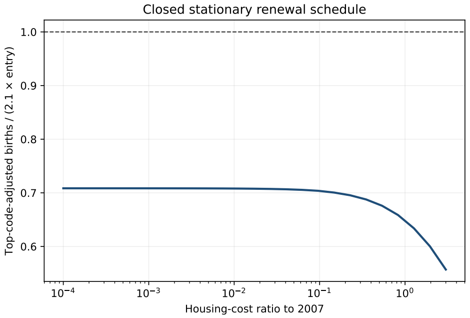


- The model misses the 2007–2011 house-price collapse and subsequent recovery.
- A fixed user-cost relation rules out independent price–rent dynamics.

The exercise isolates demographic-housing forces rather than the full housing cycle.

Sources: real Case-Shiller national house-price index and CPI rent of primary residence.

Closed-population renewal



A positive closed steady state requires $B(P)/E(P) = 1$. Over the audited price grid,

$$0.557 \leq B(P)/E(P) \leq 0.708 < 1.$$

No price restores replacement. The closed continuation is therefore a finite-horizon contraction experiment, not a transition between two positive steady states.

Rebated property tax: impact and first decade

From the same fitted 2023 distribution, permanently raise the annual property-tax rate from 1% to 2% and rebate the revenue equally. At each date, solve jointly for the house price and transfer under the maintained national supply elasticity, $\varepsilon_H = 1.75$.

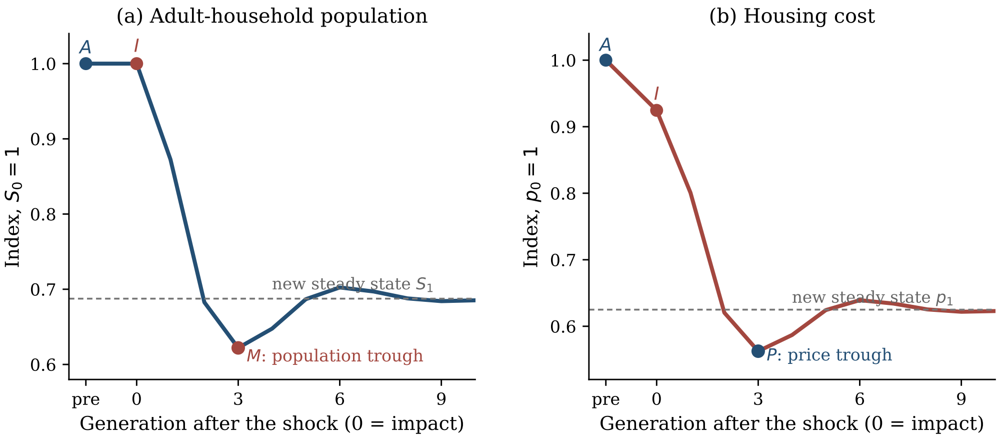
The 2031 and 2035 model dates bracket a ten-year response.

Year	ΔP	Δ user cost	Δ ownership	Δ child ownership	Δ births/adult
2023	-2.03%	$+21.61\%$	$+0.43$ pp	$+1.06$ pp	$+0.34\%$
2027	-2.46%	$+21.08\%$	$+1.35$ pp	$+2.04$ pp	$+0.38\%$
2031	-2.84%	$+20.61\%$	$+1.83$ pp	$+2.78$ pp	$+0.44\%$
2035	-3.17%	$+20.20\%$	$+2.38$ pp	$+3.43$ pp	$+0.48\%$

Capitalization persists, while the tenure response builds as household states adjust. Fertility rises modestly. Adult population is unchanged through 2035 because births first affect household entry after twenty years.

The 2023 response is robust to the alternative supply elasticities in Baum-Snow and Han (2024) and Coven et al. (2025).

Population and housing-cost dynamics

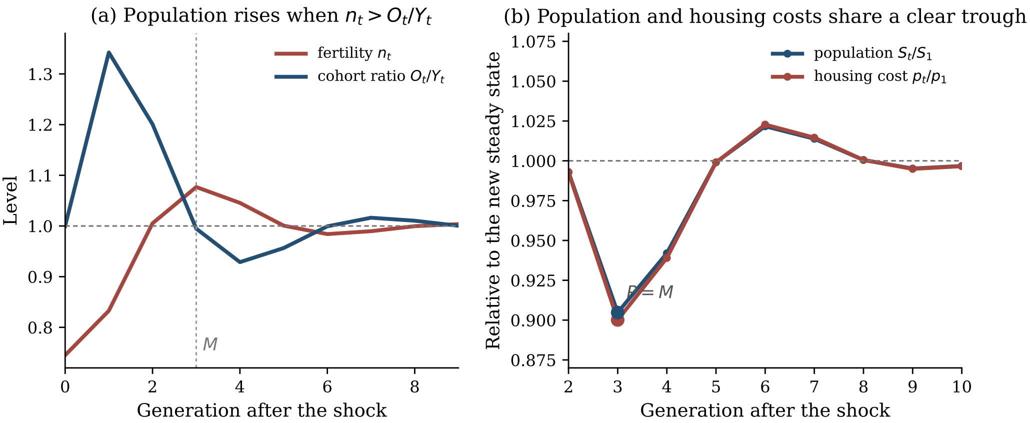


Both series reach a trough in generation 3 before converging to lower steady states.

Illustrative parameterization.

◂ Equilibrium diagram

Cohort composition and the population trough



Population falls while $n_t < O_t/Y_t$, even after $n_t > 1$. Population and price troughs coincide here but need not when housing demand differs by age.

◂ Equilibrium diagram